In-Class Lecture 12. Multivariate Analysis

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April 2023

Overview

Multivariate Normal

The p-dimensional random variable $\mathbf{X}$ with p components $(X_1, X_2, \ldots, X_p)$ with distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma}^{\mathbf{2}})$ has the following probability density function:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{p}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})}$$

Property 1

Let $\boldsymbol{X} \sim N_p(\boldsymbol{\mu}, \Sigma)$, then each component X_t for $i = 1, \ldots, p$ has a univariate normal distribution $X_i \sim N(\mu_i, \sigma_{ii})$.

Property 2

Let $\boldsymbol{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then any distinct subset of the components (say k) has a k-dimensional multivariate normal distribution.

Example. The following is a 5-dimensional multivariate normal distribution:

$$\begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{pmatrix} \sim N_5\left(\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \end{pmatrix}, \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \sigma_{14} & \sigma_{15} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} & \sigma_{24} & \sigma_{25} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} & \sigma_{34} & \sigma_{35} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} & \sigma_{44} & \sigma_{45} \\ \sigma_{51} & \sigma_{52} & \sigma_{53} & \sigma_{54} & \sigma_{55} \end{pmatrix} \right)$$

Then, the following subset is a 3-dimensional MVN distribution:

$$\begin{pmatrix} X_4 \\ X_1 \\ X_5 \end{pmatrix} \sim N_3\left(\begin{pmatrix} \mu_4 \\ \mu_1 \\ \mu_5 \end{pmatrix}, \begin{pmatrix} \sigma_{44} & \sigma_{41} & \sigma_{45} \\ \sigma_{14} & \sigma_{11} & \sigma_{15} \\ \sigma_{54} & \sigma_{51} & \sigma_{55} \end{pmatrix} \right)$$

Property 3

Recall the univariate normal property: if $X \sim N(\mu, \sigma^2)$, then cX, where c is some constant, is distributed $N(c\mu, c^2\sigma^2)$.

Let $\boldsymbol{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and let $\boldsymbol{c}' = (c_1, c_2, \ldots, c_p)$ be a fixed vector of constants, then

$$\boldsymbol{c}'\boldsymbol{X} \sim N_p(\boldsymbol{c}'\boldsymbol{\mu}, \boldsymbol{c}'\Sigma\boldsymbol{c}).$$

Property 4

Let $\boldsymbol{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and let $\boldsymbol{A}$ be a $k \times p$ matrix of fixed constants where $k \leq p$, then

$$\boldsymbol{A}\boldsymbol{X} \sim N_k(\boldsymbol{A}\boldsymbol{\mu}, \boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}').$$

Property 5

Let $\boldsymbol{X} \sim N_p(\boldsymbol{\mu}, \Sigma)$, then

$$(\boldsymbol{X} - \boldsymbol{\mu})'\Sigma^{-1}(\boldsymbol{X} - \boldsymbol{\mu}) \sim \chi^2_p.$$

Property 6

Let $\boldsymbol{X} \sim N_p(\mu, \Sigma)$, where we partition $\boldsymbol{X}$ as

$$\mathbf{X} = \begin{pmatrix} \mathbf{X_1} \\ \mathbf{X_2} \end{pmatrix}$$

where $\boldsymbol{X_1}$ is $k \times 1$ and $\boldsymbol{X_2}$ is $(p-k) \times 1$, and write

$$\begin{pmatrix} \mathbf{X_1} \\ \mathbf{X_2} \end{pmatrix} \sim N_p\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}\right)$$

where Σ_{12} is $k \times (p-k)$.

Property 6

Then the conditional distribution of $\boldsymbol{X}_1$ given $\boldsymbol{X}_2 = \boldsymbol{x}_2$, written as $(\boldsymbol{X}_1 | \boldsymbol{X}_2 = \boldsymbol{x}_2)$, is given by:

$$(\boldsymbol{X}_1 | \boldsymbol{X}_2 = \boldsymbol{x}_2) \sim N_k \left(\boldsymbol{\mu}_{1|2} (\boldsymbol{x}_2), \boldsymbol{\Sigma}_{1|2} \right)$$

$$\boldsymbol{\mu}_{1|2} (\boldsymbol{x}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\boldsymbol{x}_2 - \boldsymbol{\mu}_2)$$

$$\boldsymbol{\Sigma}_{1|2} = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21}$$

Property 6 Example

Let's look at the bivariate normal example.

$$\boldsymbol{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, Cov(\boldsymbol{X}) = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}$$

$$E(X_1|X_2 = x_2) = \mu_1 + \frac{\sigma_{12}}{\sigma_{22}}(x_2 - \mu_2)$$

$$Var(X_1|X_2 = x_2) = \sigma_{11} - \frac{{\sigma^2}_{12}}{\sigma_{22}}$$

If there are covariances between X_1 and X_2, then the conditional variance is *less*.

Example 2

Data taken from the NLSY.

Three test scores:

1. X_1=AFQT Percentile Score
2. X_2=SAT Math Score
3. X_3=SAT Verbal Score

And it looks like this:

$$\begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \sim N_3 \left(\begin{pmatrix} 67 \\ 448 \\ 409 \end{pmatrix}, \begin{pmatrix} 652 & 2485 & 2418 \\ 2485 & 14755 & 11031 \\ 2418 & 11031 & 14666 \end{pmatrix} \right)$$

Example 2

By Property 2, we can partition these matrices:

$$\begin{pmatrix} X_2 \\ X_3 \end{pmatrix} \sim N_2\left(\begin{pmatrix} 448 \\ 409 \end{pmatrix}, \begin{pmatrix} 14755 & 11031 \\ 11031 & 14666 \end{pmatrix}\right)$$

Now, we can calculate the correlation:

$$Corr\left(X_2, X_3\right) = \frac{\sigma_{23}}{\sqrt{(\sigma_{22})(\sigma_{33})}} = \frac{11031}{\sqrt{(14755)(14666)}} \approx 0.7499$$

Example 2

Now, let's look at an example conditional mean and conditional variance.

$$\left(\begin{pmatrix} X_2 \\ X_3 \end{pmatrix} | X_1 = 30\right) = \left(\begin{pmatrix} SATMath \\ SATVerbal \end{pmatrix} | AFQT = 30\right)$$

We want to know $E(X_2|X_1 = 30)$. Recall that $\mu_{2|1} = \mu_2 + \frac{\sigma_{12}}{\sigma_{11}}(x_1 - \mu_1)$. In this case, we have:

$448 + \frac{2485}{652}(30 - 67) = 308.$

Example 2

Recall the formula for conditional variance: $\sigma_{2|1} = \sigma_{22} - \frac{\sigma_{12} \cdot \sigma_{21}}{\sigma_{11}}$. In this case, we have

$$\sigma_{2|1} = 14755 - \frac{2485 \cdot 2485}{652} = 5277$$

See R script for conditional covariance calculation.

Example 2

R code.

Matrix calculations are a function in base R.

Example 2

It is interesting to note that:

$$Corr\left(SATMath, SATVerbal \,|AFQT\right) = \frac{1809}{\sqrt{(5277)(5693)}} \approx 0.3300$$

Again, this is true independent of the specific value taken on by AFQT (X_1).

In-Class Exercise

Continuing with Example 2, calculate the following quantities:

$E(X_3|X_1 = 30)$.

$\sigma_{3|1}$.

In-Class Exercise

Continuing with Example 2, calculate the following quantities:

$E(X_3|X_1 = 30) = 409 + \frac{2418}{652}(30 - 67) = 271.8$

$\sigma_{3|1} = 14666 - \frac{2418^2}{652} = 5698.6.$

Estimation and Inference

Assume $X_1, X_2, \ldots, X_n$ are normal random variables from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Then the sample mean vector and sample covariance matrix are:

$\bar{\boldsymbol{X}} = n^{-1} \sum_{i=1}^{n} \boldsymbol{X_i}$

$\boldsymbol{S} = (n-1)^{-1} \sum_{i=1}^{n} (\boldsymbol{X_i} - \bar{\boldsymbol{X}})(\boldsymbol{X_i} - \bar{\boldsymbol{X}})'$

This looks like the univariate results. These are unbiased estimators of population values (i.e. $E[\bar{\boldsymbol{X}}] = \boldsymbol{\mu}$).

Result 1

Let $X_1, X_2, \ldots, X_n$ be a random sample from a $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution. Then the following will hold:

1. $\bar{\boldsymbol{X}}$ is distributed as $N_p(\boldsymbol{\mu}, n^{-1}\boldsymbol{\Sigma})$.
2. $(n-1)\boldsymbol{S}$ is distributed as a Wishart random matrix with parameter $\boldsymbol{\Sigma}$ and degrees of freedom $n-1$.
3. $\bar{\boldsymbol{X}}$ and $\boldsymbol{S}$ are independent.

Also note that $\hat{V}(\bar{\boldsymbol{X}}) = n^{-1}\boldsymbol{S}$.

Result 2

Let $X_1, X_2, \ldots, X_n$ be a random sample from a $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution, and assume that $\boldsymbol{\Sigma}$ is known and we want to test the hypothesis H_0:$\boldsymbol{\mu} = \boldsymbol{\mu_0}$.

Then the best test rejects H_0 at the α level when

$$\left(\bar{\boldsymbol{X}} - \boldsymbol{\mu}_0\right)' \left(\tfrac{1}{n}\boldsymbol{\Sigma}\right)^{-1} \left(\bar{\boldsymbol{X}} - \boldsymbol{\mu}_0\right) \geq \chi^2_{p,\alpha}$$

We can use this fact to create confidence ellipsoids.

Example 3

Returning to the SAT Math and Verbal scores. From the 917 cases in NLSY, we estimate:

$$\bar{\boldsymbol{X}} = \begin{pmatrix} 448 \\ 409 \end{pmatrix}$$

$$\boldsymbol{S} = \begin{pmatrix} 14755 & 11031 \\ 11031 & 14666 \end{pmatrix}$$

Example 3

$$n^{-1}\mathbf{S} = \begin{pmatrix} 16.0905 & 12.0294 \\ 12.0294 & 15.9935 \end{pmatrix}$$

We want to test the hypothesis:

$$H_0 : \begin{pmatrix} \mu_{Math} \\ \mu_{Verbal} \end{pmatrix} = \begin{pmatrix} 450 \\ 400 \end{pmatrix} \text{ versus}$$

$$H_A : \begin{pmatrix} \mu_{Math} \\ \mu_{Verbal} \end{pmatrix} \neq \begin{pmatrix} 450 \\ 400 \end{pmatrix}.$$

Example 3

$$\left(\begin{pmatrix} 448 \\ 409 \end{pmatrix} - \begin{pmatrix} 450 \\ 400 \end{pmatrix}\right)' \begin{pmatrix} 16.091 & 12.029 \\ 12.029 & 15.994 \end{pmatrix}^{-1} \left(\begin{pmatrix} 448 \\ 409 \end{pmatrix} - \begin{pmatrix} 450 \\ 400 \end{pmatrix}\right) = ?$$

Solve for the unknown value, which has a χ^2 distribution.

Example 3

$$\left(\begin{pmatrix} 448 \\ 409 \end{pmatrix} - \begin{pmatrix} 450 \\ 400 \end{pmatrix}\right)' \begin{pmatrix} 16.091 & 12.029 \\ 12.029 & 15.994 \end{pmatrix}^{-1} \left(\begin{pmatrix} 448 \\ 409 \end{pmatrix} - \begin{pmatrix} 450 \\ 400 \end{pmatrix}\right) = 15.98$$

Now check the χ^2 distribution. We see that $\chi^2_{2,0.05} = 5.99$. Our value is much greater than that, so we reject H_0.

Example 3

Now let's invert that procedure. We have a χ^2 value and want to see which $\boldsymbol{\mu_0}$ values are implied by this value.

$$\left(\begin{pmatrix} 448 \\ 409 \end{pmatrix} - \begin{pmatrix} \mu_{Math} \\ \mu_{Verbal} \end{pmatrix}\right)' \begin{pmatrix} 16.0905 & 12.0294 \\ 12.0294 & 15.9935 \end{pmatrix}^{-1} \left(\begin{pmatrix} 448 \\ 409 \end{pmatrix} - \begin{pmatrix} \mu_{Math} \\ \mu_{Verbal} \end{pmatrix}\right) = 5.99$$

Now, solve for the missing $\boldsymbol{\mu_0}$ values. The solution creates an ellipse.

Example 3

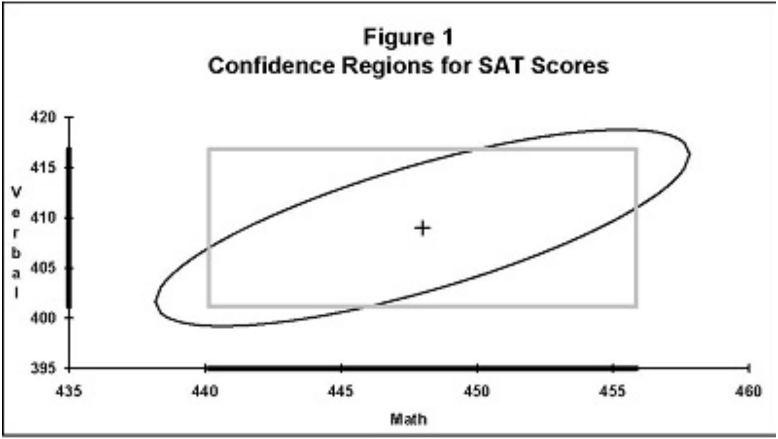
Figure 1
Confidence Regions for SAT Scores
Verbal
420
415
410
405
400
395
435
440
445
450
455
460
Math

Problem

Recall the univariate situation. When we replaced a known variance in the denominator of the Z statistic with a sample estimate, we had a random variable with a t-distribution.

The same thing happens when looking at the multivariate situation. When we use the sample estimate of a covariance matrix, then the earlier test is inappropriate (result 2).

Result 3

Let $X_1, X_2, \ldots, X_n$ be a random sample from a $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution, and assume that $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are both unknown and we want to test the hypothesis H_0:$\boldsymbol{\mu} = \boldsymbol{\mu_0}$.

Hotelling's T^2 Statistic:
$T^2 = \left(\bar{\boldsymbol{X}} - \boldsymbol{\mu}_0\right)' \left(\frac{1}{n}\boldsymbol{S}\right)^{-1} \left(\bar{\boldsymbol{X}} - \boldsymbol{\mu}_0\right)$

Then reject H_0 when:
$T^2 > \frac{(n-1)p}{(n-p)} F^{p}_{n-p}(\alpha)$

Example 4

Recall our hypothesis:

$$H_0 : \begin{pmatrix} \mu_{Math} \\ \mu_{Verbal} \end{pmatrix} = \begin{pmatrix} 450 \\ 400 \end{pmatrix} \text{ versus}$$

$$H_A : \begin{pmatrix} \mu_{Math} \\ \mu_{Verbal} \end{pmatrix} \neq \begin{pmatrix} 450 \\ 400 \end{pmatrix}.$$

Example 4

Calculate Hotelling's T^2:

$$T^2 = (\bar{\boldsymbol{X}} - \mu_0)' \left(\frac{1}{n}\boldsymbol{S}\right)^{-1} (\bar{\boldsymbol{X}} - \mu_0)$$

$$T^2 = (917)\left(\begin{pmatrix} 448 \\ 409 \end{pmatrix} - \begin{pmatrix} 450 \\ 400 \end{pmatrix}\right)' \begin{pmatrix} 14755 & 11031 \\ 11031 & 14666 \end{pmatrix}^{-1} \left(\begin{pmatrix} 448 \\ 409 \end{pmatrix} - \begin{pmatrix} 450 \\ 400 \end{pmatrix}\right) = 15.98$$

Now compare that to the appropriate value from the reference distribution:

$$\frac{(n-1)p}{(n-p)} F^p_{n-p}(0.05) = \frac{(916)2}{915}(3.00) = 6.01$$

$15.98 > 6.01$. Therefore, reject H_0.

Convergence

Now, you might be thinking that it looks a lot like the prior example.

This is because there is an asymptotic convergence: $\frac{(n-1)p}{(n-p)} F^p_{n-p} \to \chi^2_p$ as $n \to \infty$.

Alternative Motivation

First, if we multiply our multivariate normal by a vector of constants, we have:

$$\boldsymbol{c}'\bar{\boldsymbol{X}} = \sum_{i=1}^{p} c_i \bar{X}_i.$$

From the properties of expected value and variance we know:

$$E\left\{\boldsymbol{c}'\bar{\boldsymbol{X}}\right\} = \boldsymbol{c}'\boldsymbol{\mu}$$

$$\hat{V}\left\{\boldsymbol{c}'\bar{\boldsymbol{X}}\right\} = n^{-1}\mathbf{c}'\mathbf{S}\mathbf{c}$$

Alternative Motivation

Now form the univariate t-statistic:

$$t_c = \frac{\mathbf{c}'(\bar{\mathbf{X}} - \boldsymbol{\mu})}{\sqrt{n^{-1}\mathbf{c}'\mathbf{S}\mathbf{c}}}$$

Square both sides of the equation:

$$t_c^2 = \frac{\mathbf{c}'(\bar{\mathbf{X}} - \boldsymbol{\mu})'(\bar{\mathbf{X}} - \boldsymbol{\mu})\mathbf{c}}{n^{-1}\mathbf{c}'\mathbf{S}\mathbf{c}}$$

Then it can be shown that:

$$\max_{c \in \Re^p} \left(t_c^2\right) = T^2 = n(\bar{\mathbf{X}} - \boldsymbol{\mu})'\mathbf{S}^{-1}\left(\bar{\mathbf{X}} - \boldsymbol{\mu}\right)$$

In words, the Hotelling T^2 is the largest squared univariate t-statistic that can be constructed from the data.

Result 4

Let $X_1, X_2, \ldots, X_n$ be a random sample from a $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution, and assume that $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are both unknown, and let $\boldsymbol{c}$ be any p-dimensional column vector. Then simultaneously for all values of c in $\Re^p$, the confidence interval:

$$\left[\boldsymbol{c}'\bar{\boldsymbol{X}} - \sqrt{\tfrac{(n-1)p}{(n-p)} F^p_{n-p}(\alpha)}\sqrt{n^{-1}\boldsymbol{c}'\boldsymbol{S}\boldsymbol{c}}, \boldsymbol{c}'\bar{\boldsymbol{X}} + \sqrt{\tfrac{(n-1)p}{(n-1)} F^p_{n-p}(\alpha)}\sqrt{n^{-1}\boldsymbol{c}'\boldsymbol{S}\boldsymbol{c}}\right]$$

will contain $\boldsymbol{c}'\boldsymbol{\mu}$ with probability $1-\alpha$. These are called simultaneous T^2 confidence intervals.

Example 5

Back to the same data:

$$\begin{aligned} \bar{X}_{math} &= 448 \\ \bar{X}_{verbal} &= 409 \\ \hat{V}(\bar{X}_{math}) &= 16.0905 \\ \hat{V}(\bar{X}_{verbal}) &= 15.9935 \end{aligned}$$

For these data we can easily construct univariate 95% confidence intervals.

For μ_{math} the interval is $[440.138, 455.862]$.
For μ_{verbal} the interval is $[401.162, 416.838]$.

Example 5

But we know that these variables are related (bivariate normal). Let's estimate the simultaneous T^2 confidence intervals using:

$$\left[\boldsymbol{c}'\bar{\boldsymbol{X}} - \sqrt{\tfrac{(n-1)p}{(n-p)} F^{p}_{n-p}(\alpha)} \sqrt{n^{-1}\boldsymbol{c}'\boldsymbol{S}\boldsymbol{c}}, \boldsymbol{c}'\bar{\boldsymbol{X}} + \sqrt{\tfrac{(n-1)p}{(n-1)} F^{p}_{n-p}(\alpha)} \sqrt{n^{-1}\boldsymbol{c}'\boldsymbol{S}\boldsymbol{c}}\right]$$

For μ_{math} the interval is $[438.1825, 457.8174]$.
For μ_{verbal} the interval is $[399.2122, 418.7878]$.

These intervals are slightly wider.

Example 5

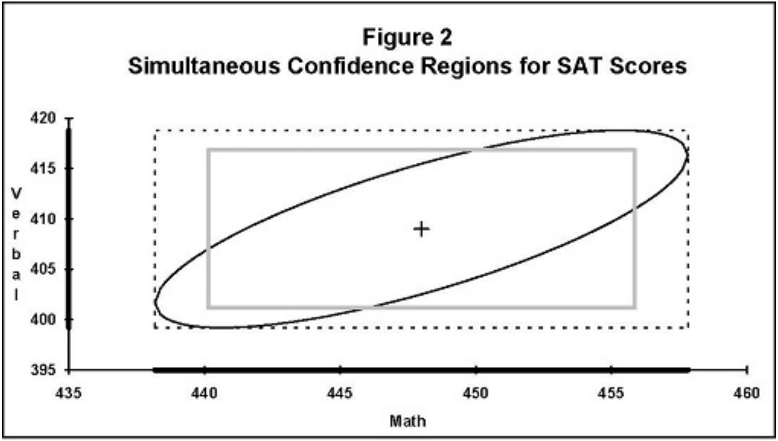
Figure 2
Simultaneous Confidence Regions for SAT Scores
420
415
410
405
400
395
Verbal
435
440
445
450
455
460
Math

Pooled Tests

We may want to test the difference in mean vectors $(\boldsymbol{\mu_1} - \boldsymbol{\mu_2})$ between two multivariate normal distributions.

Let $X_1, X_2, \ldots, X_{n1}$ be a random sample of size n_1 drawn from a $N_p(\boldsymbol{\mu_1}, \Sigma)$ distribution.

Let $Y_1, Y_2, \ldots, Y_{n2}$ be a random sample of size n_2 drawn from a $N_p(\boldsymbol{\mu_2}, \Sigma)$ distribution.

NOTE: The covariance matrices are the same in the two distributions. We need this to be true for this to work!

Pooled Tests

Our hypothesis is H_0:$\boldsymbol{\mu_1} - \boldsymbol{\mu_2} = \boldsymbol{\delta}$.

Recall that $\boldsymbol{S_2} = (n_2 - 1)\sum_{i=1}^{n_2}(\boldsymbol{Y_i} - \bar{\boldsymbol{Y}})((\boldsymbol{Y_i} - \bar{\boldsymbol{Y}})'$.

If we can assume that $\boldsymbol{S_1}$ and $\boldsymbol{S_2}$ come from the same distribution, then we can "pool" them to increase the power of our test:

$\boldsymbol{S}_{Pooled} = \frac{(n_1-1)\boldsymbol{S}_1+(n_2-1)\boldsymbol{S}_2}{(n_1+n_2-2)}$

Pooled Tests

Now we can calculate the T^2 statistic:

$$T^2 = (\bar{\boldsymbol{X}} - \bar{\boldsymbol{Y}} - \delta)' \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \boldsymbol{S}_{Pooled} \right]^{-1} (\bar{\boldsymbol{X}} - \bar{\boldsymbol{Y}} - \delta)$$

We reject H_0 if:

$$T^2 > \frac{(n_1+n_2-2)p}{(n_1+n_2-p-1)} F^p_{n_1+n_2-p-1}(\alpha)$$

Example 6

Back to the same data.

$X_1, X_2, \ldots, X_{407}$ is a random sample of 407 men with a $N_3(\boldsymbol{\mu_1}, \boldsymbol{\Sigma})$ distribution.

$Y_1, Y_2, \ldots, Y_{510}$ is a random sample of 510 women with a $N_3(\boldsymbol{\mu_2}, \boldsymbol{\Sigma})$ distribution.

Test the null hypothesis: H_0:$\boldsymbol{\mu_1} = \boldsymbol{\mu_2}$.

Example 6

$$\bar{\boldsymbol{X}} = \begin{pmatrix} 72.412776 \\ 484.69287 \\ 425.40541 \end{pmatrix}$$

$$\bar{\boldsymbol{Y}} = \begin{pmatrix} 62.186275 \\ 418.80392 \\ 396.45098 \end{pmatrix}$$

Example 6

$$\boldsymbol{S}_1 = \begin{pmatrix} 570.58782 & 2331.9744 & 2298.0835 \\ 2331.9744 & 15281.125 & 11725.802 \\ 2298.0835 & 11725.802 & 15422.434 \end{pmatrix}$$

$$\boldsymbol{S}_2 = \begin{pmatrix} 671.18134 & 2312.8558 & 2387.185 \\ 2312.8558 & 12433.341 & 9650.1672 \\ 2387.185 & 9650.1672 & 13718.421 \end{pmatrix}$$

$$\boldsymbol{S}_{Pooled} = \begin{pmatrix} 626.5464 & 2321.339 & 2347.6492 \\ 2321.339 & 13696.948 & 10571.159 \\ 2347.649 & 10571.159 & 14474.518 \end{pmatrix}$$

Example 6

$T^2 = 90.13.$

$\frac{(n_1+n_2-2)p}{(n_1+n_2-p-1)} F^p_{n_1+n_2-p-1}(0.05) = \frac{(915)3}{(913)}(2.61) = 7.85$

Example 6

See R program (work from mean vectors, covariance matrices).

Just as with a two-sample t-test, we should test for equality of variances.

Example 6

In R, use the `DescTools` package.

```
library(DescTools)
HotellingsT2Test(xscores, y = yscores)
```